Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/26/2018
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Investigate Monocot Plant Adaptation to Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	09/12/2014	**End Date:**	09/15/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS (APEX-06) NOTE: End date changed to 09/15/2019 per NSSC information (Ed., 3/12/19) NOTE: End date changed to 3/11/2019 per NSSC information (Ed., 9/14/18) NOTE: End date changed to 9/11/2018 per NSSC information (Ed., 12/13/17) NOTE: End date changed to 9/11/2017 per NSSC information (Ed., 6/14/16)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AT23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Most major cereal grain crops are monocots. Yet, most investigations of plant adaptation to the spaceflight environment have been carried out on the dicotyledonous model plant Arabidopsis thaliana. It remains unknown whether the conclusions from such studies can be extrapolated to monocotyledonous plants. To fill this knowledge gap, we propose to investigate the growth, development, and transcriptome profiles of Brachypodium distachyon seedlings under spaceflight conditions, and compare these adaptive responses to those recently documented for Arabidopsis seedlings. These investigations will lead to a better understanding of the molecular and developmental mechanisms that contribute to Brachypodium adaptation to spaceflight conditions. In the long term, these data may also lead to the development of better mitigation strategies aimed at improving monocot adaptability to spaceflight parameters.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Because most cultivated crops are monocotyledonous plants, there is tremendous interest in better understanding the molecular mechanisms that govern their growth behaviors and adaptation to stress. In addition to contributing to a better understanding of the molecular mechanisms that modulate plant adaptation to the microgravity environment, an important step toward engineering crops that are better suited to serve as components of the bioregenerated life-support systems during spaceflight, our work with Brachypodium distachyon should provide new insights into the molecular mechanisms that control root growth and development. Because root growth and its architecture have recently been recognized as key contributors to plant productivity and adaptation to environmental stress, our research is also likely to have long-term impact in agriculture, horticulture, and forestry. Furthermore, many of the behaviors under investigation involve auxin-related processes. This plant hormone is a key contributor to most aspects of plant growth, development, and response to the environment. Therefore, this project is likely to yield important new insights into some of the key mechanisms that control plant growth, development, and response to biotic and abiotic stresses.
Task Progress:	As summarized last year, we developed a novel plant growth vessel (named APEX Growth Chamber) to support the development of a monocot model plant similar to agricultural crops, Brachypodium distachyon, and investigate its adaptation to the microgravity conditions encountered during spaceflight as well as in the International Space Station. This work was performed by Dr. Shih-Heng Su in collaboration with Dr. Howard Levine (NASA Space Biology Program) and the Kennedy Space Center (KSC) Support Team lead by Anne Marie Campbell and Shawn Stephens. This chamber was designed as a modification of an Oasis foam-based growth device previously developed by Dr. Howard Levine for plant growth under microgravity. It can be used to grow up to 24 Brachypodium seedlings for a period of 4 days, allowing both shoots and roots to remain on the surface of the foam support, visible to the lens of a camera and accessible for easy harvesting at the end of the experiment. This device allows experimentalists to insert surface-sterilized dry seeds into a dry foam block before launch, and keep the seeds dry, hence dormant, during spaceflight until the unit reaches the International Space Station (ISS). At that time, the experiment can be activated by injection of a nutrient solution into the foam, thereby imbibing the seeds. The following germination can be optimized / synchronized by red light treatment over a period of 24 hours. Subsequently, green and blue lights can be turned on, and the seedlings are allowed to grow for three more days, giving plant material that can be photographed for biometric analysis, and harvested, then fixed, into Kennedy Fixation Tubes, using RNA later. Fixed materials can then be returned to Earth for RNA extraction and expression characterization. This year, we tested and refined this novel experimental protocol, which successfully passed a Science Verification Test (SVT) at Kennedy Space Center (fall of 2017), and an Experiment Verification Test (EVT) in January of 2018. This experiment uses three Brachypodium distachyon accessions known to display distinct root growth behaviors under normal gravity conditions: Bd21 (whose genome has served as a reference for Brachypodium), BD21-3 (which has been optimized for use in transformation experiments), and Gaz-8 (which displays different root growth behaviors relative to Bd21 and Bd21-3 when grown on the ground). These tests successfully demonstrated high germination rates for all three accessions, sufficient seedling organ growth to allow subsequent molecular characterization of expression profiles, and ability to extract good quality RNA for subsequent to RT-qPCR analysis. Therefore, this project was given the green light to proceed to its implementation phase on ISS, under the name 'APEX-06'. The material was prepared for flight on March 27-30, 2018, and launched on SpaceX-14 on April 2, 2018. One astronaut activated the experiment on ISS by injecting growth medium into the 12 APEX growth chambers (4 repeats for each of the three accessions described above) on April 12, 2018. Seeds were allowed to germinate in the presence of red light for 24 hours, then blue and green light were added for an additional 3 days. At the end of this growth period, astronaut Dr. Tingle recovered the APEX growth chambers from VEGGIE, took photographs of the developing seedlings on each face of the foam block, harvested the seedlings, and placed them into Kennedy Fixation Tubes (KFT) for fixation with RNA later. Most APEX Growth Chambers contained a few seedlings that displayed an unusual growth pattern with roots detached from the foam surface, therefore showing evidence of stress. These seedlings were harvested separately from those whose root tips were still contacting the block surface. Fixation was allowed to proceed for a period of 24 h, at which time the KFTs were transferred to cold storage (-80°C) and returned to the ground for molecular analysis in the Principal Investigator's laboratory. A parallel time-delayed ground-based control experiment was carried out at KSC, during which the environmental conditions experienced by the plants during their growth period in VEGGIE on ISS were recapitulated in plant growth units at KSC. At the end of this experiment, the seedlings were also photographed, harvested, and fixed in RNA later for subsequent RNA extraction and transcriptomic analysis. Photographs taken during both the flight experiment on ISS and the ground-based control experiment at KSC were used to investigate seedling organs growth under microgravity or control 1 g conditions, revealing a slight alteration in the rate of growth under microgravity. To better understand the molecular mechanisms that underlie Brachypodium seedling organs adaptation to the microgravity environment, fixed frozen seedlings will be dissected to separate shoots and roots. RNA will be extracted from these tissues and used in RNA-seq experiments aimed at comparing organ-specific expression profiles between microgravity-grown and control samples. A second ground-based control experiment was also carried out at KSC by Dr. Jeffrey Richards in July 2018, to investigate the molecular mechanisms that modulate root and shoot responses to gravistimulation and mechanostimulation under 1g conditions. In this experiment, Bd21 seedlings were germinated and grown in APEX

	growth chambers under the same conditions as those experienced during the APEX-06 mission on ISS. After 4 days of growth, the APEX growth units were either gravistimulated by a 180° reorientation, mechanostimulated by a 360° stimulus, or unstimulated (control). Four APEX growth chambers were subjected to each one of these treatments (four biological repeats). After 5 min of stimulation, the seedlings were harvested from the chambers and fixed in RNA later, then frozen at -80°C. Fixed seedlings will be dissected to separate shoots from roots. RNA will then be extracted from these tissues, and used in an RNAseq analysis to characterize their transcriptomic profiles. Groups of genes found to be differentially expressed under microgravity in ISS relative to the first ground-based control will be compared to those found to be differentially expressed in response to gravistimulation under 1 g or in response to mechanostimulation (as determined in the second ground-based control). Groups of genes that are found to be differentially expressed under both microgravity and gravistimulation conditions will be retained for functional analysis as they are more likely to directly contribute to gravity signal transduction. In addition to the APEX-06 experiment, we have also continued our analysis of Brachypodium root gravitropism and oscillatory root growth behavior on agar surfaces. Last year, we reported using the natural variation that exists between 42 Brachypodium accessions to investigate the molecular mechanisms that underlie the variation in kinetics of root gravitropism and properties of oscillatory root growth. This year, we have extended this analysis to include more Brachypodium accessions and a larger number of seedlings per accession, making the identification of associated loci by Genome-Wide Association Study (GWAS) more robust and meaningful. Importantly, this analysis allowed the identification of several polymorphisms associated with quantitative parameters that recapitulate different aspects of root-tip oscillatory growth. Some of the significant SNPs identified in this work mapped to genes that encode a sugar-binding protein kinase and a polar auxin transporter. These genes' annotations suggest roles in proprioception and auxin transport/signaling, respectively, and experiments are underway to verify their contribution to the control of oscillatory root growth. Overall, these studies should improve our understanding of the molecular mechanisms that control coordinated cell expansion in roots and its roles in growth regulation, morphogenesis, and growth responses to gravity and/or microgravity. These studies may impact our ability to engineer plants that are better suited for use in the bioregenerative life-support systems that will accompany long-term space exploration missions, or more simply for effective growth and productivity in natural and/or agricultural ecosystems on Earth.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Gibbs NM, Rouhana LV, Masson PH. "Quantitative trait loci for root growth response to cadaverine in Arabidopsis." Methods Mol Biol. 2018;1694:233-56. Epub 2017 Oct 28. https://doi.org/10.1007/978-1-4939-7398-9_22 ; PubMed PMID: 29080172 , Jan-2018